Tiny Snakes Roll Four Times Faster Than They Slither

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Imagine a tiny snake that does not slither away from danger. It rolls. Dwarf reedsnakes in Southeast Asia can roll at about four times their usual slithering speed, and scientists have worked out how they pull off the trick.

Rolling sounds simple, but for an animal it is surprisingly hard. A wheel spins around a fixed **axle** while the part it carries stays still. An animal has no such trick. If its whole body spins, its head spins too, and that makes balance, steering, and keeping track of direction much harder. The faster the roll, the worse the problem gets.

Dwarf reedsnakes, members of the **genus** Pseudorabdion, are among the few animals known to roll on purpose to escape **predators**. The behaviour had been noted in field guides and seen in the field before, but researchers had not fully understood the movements behind it.

To find out, they filmed the snakes from several angles. While rolling, the snakes reached peak speeds of 1.72 to 3.69 metres per second, about four times their fastest slithering speed. Even at that pace, they stayed on a racetrack only 20 centimetres wide, so this is not wild tumbling.

The secret is a carefully timed pattern. Instead of spinning their whole bodies at once, the snakes alternate between rolling on one side and swinging a curved section of the body downward beside the head. Researchers called that downward sweep a **stroke**. With each cycle, the side touching the ground switches, and the **stroke** changes direction, left to right, then right to left.

Best of all, the head stays out of the spin. While the rear of the body **rotates** rapidly, the head points forward at a nearly constant height, like a steady lookout. That control turns a clumsy tumble into a speedy escape.

Word Treasure

axle -- A rod that a wheel turns around.
genus -- A group of closely related living things.
predators -- Animals that hunt other animals for food.
stroke -- Here, a downward sweep of part of the snake's body.
rotates -- Turns around a central point.

Background you need

Snakes usually move by sliding their bodies along the ground, a movement called slithering.

Animals that hunt them are called predators, and escaping quickly can mean survival.

Rolling is a very different way to move, because a rolling body must keep its balance and direction while it turns.

Researchers study such movements by filming animals from several angles, so they can see exactly how each part of the body moves.

Break it down

LEAD: Dwarf reedsnakes roll to escape danger, about four times faster than they slither
KEY EVENT: Scientists worked out how the snakes pull off the trick
+- WHO: Dwarf reedsnakes, members of the genus Pseudorabdion, in Southeast Asia
+- HOW: Filmed from several angles
+- EVIDENCE: Peak rolling speeds of 1.72 to 3.69 metres per second, staying on a 20-centimetre-wide racetrack
+- SECRET: Alternating sides and a stroke, with the head staying steady
CONCLUSION: Controlled rolling turns a clumsy tumble into a speedy escape

Why it matters

It shows how animals can solve a hard movement problem with timing and control, and it helps scientists understand how bodies, balance and steering work in nature.

Test what you remember

1. How much faster can dwarf reedsnakes roll than they can slither?

(A) About four times faster
(B) About twice as fast
(C) About ten times faster
(D) At about the same speed

2. What peak rolling speeds did the snakes reach?

(A) 0.5 to 1.0 metres per second
(B) 5 to 6 metres per second
(C) 20 to 25 metres per second
(D) 1.72 to 3.69 metres per second

3. How wide was the racetrack the rolling snakes stayed on?

(A) 50 centimetres
(B) 20 centimetres
(C) 10 centimetres
(D) 5 centimetres

4. In this story, what is a stroke?

(A) A mark left on the ground
(B) A downward sweep of a curved section of the body beside the head
(C) A sound the snake makes
(D) A type of snake bite

5. What part of the snake stays out of the spin while it rolls?

(A) The head
(B) The middle of the body
(C) The tail
(D) The whole body

6. Why do dwarf reedsnakes roll?

(A) To find food
(B) To escape predators
(C) To keep warm
(D) To attract a mate

Answer key (try the quiz first!): 1: A 2: D 3: B 4: B 5: A 6: B

Questions to think about

MY THOUGHTS (at least 20 words):